

Polar Coordinates & Polar Graphs

ANSWER KEY



Converting between polar and rectangular coordinates

- 1 Convert the polar point $\left(4, \frac{2\pi}{3}\right)$ to rectangular coordinates.

$$x = r \cos \theta = 4 \cos \left(\frac{2\pi}{3}\right) = 4 \left(-\frac{1}{2}\right) = -2. \quad y = r \sin \theta = 4 \sin \left(\frac{2\pi}{3}\right) = 4 \left(\frac{\sqrt{3}}{2}\right) = 2\sqrt{3}. \quad \text{Rectangular point: } (-2, 2\sqrt{3}).$$

- 2 Convert the rectangular point $(-3, 3)$ to polar coordinates with $r > 0$ and $0 \leq \theta < 2\pi$.

$$r = \sqrt{(-3)^2 + 3^2} = \sqrt{18} = 3\sqrt{2}. \quad \text{Since } (-3, 3) \text{ is in Quadrant II, } \theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}. \quad \text{Polar point: } \left(3\sqrt{2}, \frac{3\pi}{4}\right).$$

Converting polar equations to rectangular form

- 3 Convert the polar equation $r = 4 \cos \theta$ to rectangular form, and identify the curve.

Multiply both sides by r : $r^2 = 4r \cos \theta$, so $x^2 + y^2 = 4x$. Completing the square: $(x - 2)^2 + y^2 = 4$ — a circle of radius 2 centered at $(2, 0)$.

- 4 Convert the rectangular equation $x^2 + y^2 = 9$ to polar form.

Since $r^2 = x^2 + y^2$, the equation becomes $r^2 = 9$, so $r = 3$ — a circle of radius 3 centered at the pole.

Visualizing a rectangular (r vs. θ) graph of a polar rose

- 5 The graph shows $r = 3 \cos(2\theta)$ plotted in rectangular form, with θ on the horizontal axis (in radians) and r on the vertical axis — a standard technique for analyzing a polar rose curve before sketching it.

a State how many petals the actual polar rose $r = 3 \cos(2\theta)$ has, using the rule for $r = a \cos(n\theta)$ with even n . When n is even, the rose $r = a \cos(n\theta)$ has $2n$ petals; here $n = 2$ gives 4 petals.

b Read off the graph: at which values of θ in $[0, 2\pi)$ does $r = 0$ (these are the angles where the rose curve passes through the pole)? $r = 0$ when $\cos(2\theta) = 0$, i.e. $2\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$, so $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ — matching the four zero-crossings shown.

Symmetry and special polar curves

- 6 Identify the type of curve represented by $r = 2 + 2 \sin \theta$ and state its key feature.

This is a cardioid (heart-shaped curve), since it has the form $r = a + a \sin \theta$ (or $\cos \theta$) with equal constant and coefficient. It passes through the pole when $\sin \theta = -1$, i.e. $\theta = \frac{3\pi}{2}$.

- 7 Determine whether the graph of $r = 3\cos\theta$ is symmetric about the polar axis (the x -axis), and justify your answer.

Replacing θ with $-\theta$: $r = 3\cos(-\theta) = 3\cos\theta$, unchanged. So the graph is symmetric about the polar axis.

Multiple representations of the same polar point

- 8 Give two other polar representations of the point $\left(5, \frac{\pi}{4}\right)$: one with a negative r -value, and one with a different positive angle (adding 2π).

With negative r : $\left(-5, \frac{\pi}{4} + \pi\right) = \left(-5, \frac{5\pi}{4}\right)$. With added revolution: $\left(5, \frac{\pi}{4} + 2\pi\right) = \left(5, \frac{9\pi}{4}\right)$.

Applications of polar curves

- 9 A radar station models the range of its signal by the polar curve $r = 6 + 2\cos\theta$ (in km), where θ is measured from due east. Find the maximum and minimum detection range, and the direction of each.

Maximum range: $r = 6 + 2(1) = 8$ km, at $\theta = 0$ (due east). Minimum range: $r = 6 + 2(-1) = 4$ km, at $\theta = \pi$ (due west).

Distance between points in polar form

- 10 Find the distance between the polar points $\left(3, \frac{\pi}{6}\right)$ and $\left(5, \frac{2\pi}{3}\right)$ by first converting both to rectangular coordinates.

Point 1: $x = 3\cos\left(\frac{\pi}{6}\right) = 3\left(\frac{\sqrt{3}}{2}\right) = \frac{3\sqrt{3}}{2}$, $y = 3\sin\left(\frac{\pi}{6}\right) = \frac{3}{2}$. Point 2: $x = 5\cos\left(\frac{2\pi}{3}\right) = -\frac{5}{2}$, $y = 5\sin\left(\frac{2\pi}{3}\right) = \frac{5\sqrt{3}}{2}$. Distance $\approx \sqrt{(2.60 - (-2.5))^2 + (1.5 - 4.33)^2} \approx \sqrt{26.0 + 8.01} \approx 5.83$.

Identifying polar curve families

- 11 Classify each polar equation as a circle, cardioid, limaçon (with inner loop), or rose, and briefly justify: (a) $r = 5$, (b) $r = 2 + 4\sin\theta$, (c) $r = 4\sin(3\theta)$.

(a) $r = 5$ is a circle of radius 5 centered at the pole (constant r). (b) $r = 2 + 4\sin\theta$ has $|a| < |b|$ (here $a = 2 < b = 4$), so it is a limaçon with an inner loop. (c) $r = 4\sin(3\theta)$ has the rose form $r = a\sin(n\theta)$ with odd $n = 3$, giving a 3-petaled rose.

Points of intersection of polar curves

- 12 Find the values of θ in $[0, 2\pi)$ where the polar curves $r = 2$ and $r = 4\cos \theta$ intersect.

Set $2 = 4\cos \theta$: $\cos \theta = \frac{1}{2}$, giving $\theta = \frac{\pi}{3}$ or $\theta = \frac{5\pi}{3}$ (also written $-\frac{\pi}{3}$).

Area in polar coordinates (introductory)

- 13 Explain, without evaluating an integral, why the polar area formula $A = \frac{1}{2} \int_a^b r^2 d\theta$ uses r^2 rather than r , by comparing it to the area of a circular sector $A = \frac{1}{2} r^2 \theta$.

A circular sector of radius r and angle θ has area $\frac{1}{2} r^2 \theta$ — note the r^2 , since area scales with the square of the radius. The polar area formula generalizes this by summing infinitely many thin sectors, each with area $\frac{1}{2} r^2 d\theta$, which is exactly why r (not r^2 or r^1 alone without justification) appears squared in the integrand.

- 14 A circular sector has radius 6 and central angle $\frac{\pi}{3}$. Use the sector-area formula $A = \frac{1}{2} r^2 \theta$ to find its area.

$$A = \frac{1}{2} (6)^2 \left(\frac{\pi}{3} \right) = \frac{1}{2} (36) \left(\frac{\pi}{3} \right) = 6\pi \approx 18.85.$$

Multiple choice — easy 15

- 15 Classify the polar curve $r = 3\sin \theta$.

- a. Circle through the pole ✓
- b. Limaçon with an inner loop
- c. Cardioid
- d. Limaçon without an inner loop

For $r = 3\sin \theta$: this is a circle through the pole.

- 16 Classify the polar curve $r = 7 + 5\cos \theta$.

- a. Cardioid
- b. Circle through the pole
- c. Limaçon with an inner loop
- d. Limaçon without an inner loop ✓

For $r = 7 + 5\cos \theta$: this is a limaçon without an inner loop.

17 Classify the polar curve $r = 2 + 5\cos \theta$.

a. Limaçon with an inner loop ✓

b. Circle through the pole

c. Limaçon without an inner loop

d. Cardioid

For $r = 2 + 5\cos \theta$: this is a limaçon with an inner loop.

18 Classify the polar curve $r = 4 + 6\sin \theta$.

a. Limaçon with an inner loop ✓

b. Cardioid

c. Limaçon without an inner loop

d. Circle through the pole

For $r = 4 + 6\sin \theta$: this is a limaçon with an inner loop.

19 Classify the polar curve $r = 1 + 4\cos \theta$.

a. Limaçon without an inner loop

b. Circle through the pole

c. Limaçon with an inner loop ✓

d. Cardioid

For $r = 1 + 4\cos \theta$: this is a limaçon with an inner loop.

20 Classify the polar curve $r = 8\cos \theta$.

a. Limaçon with an inner loop

b. Circle through the pole ✓

c. Cardioid

d. Limaçon without an inner loop

For $r = 8\cos \theta$: this is a circle through the pole.

21 Classify the polar curve $r = 1 + 5\sin \theta$.

- a. Limaçon without an inner loop
- b. Limaçon with an inner loop ✓**
- c. Cardioid
- d. Circle through the pole

For $r = 1 + 5\sin \theta$: this is a limaçon with an inner loop.

22 Classify the polar curve $r = 3 + 8\cos \theta$.

- a. Circle through the pole
- b. Limaçon with an inner loop ✓**
- c. Cardioid
- d. Limaçon without an inner loop

For $r = 3 + 8\cos \theta$: this is a limaçon with an inner loop.

23 Classify the polar curve $r = 2 + 2\cos \theta$.

- a. Circle through the pole
- b. Cardioid ✓**
- c. Limaçon with an inner loop
- d. Limaçon without an inner loop

For $r = 2 + 2\cos \theta$: this is a cardioid.

24 Classify the polar curve $r = 3 + 3\sin \theta$.

- a. **Cardioid ✓**
- b. Limaçon without an inner loop
- c. Limaçon with an inner loop
- d. Circle through the pole

For $r = 3 + 3\sin \theta$: this is a cardioid.

25 Classify the polar curve $r = 7\cos \theta$.

- a. Cardioid
- b. Limaçon with an inner loop
- c. Limaçon without an inner loop
- d. **Circle through the pole ✓**

For $r = 7\cos \theta$: this is a circle through the pole.

26 Classify the polar curve $r = 3\sin(5\theta)$.

- a. **Rose with 5 petals ✓**
- b. Circle through the pole
- c. Cardioid
- d. Rose with 12 petals

For $r = 3\sin(5\theta)$: this is a rose with 5 petals.

27 Classify the polar curve $r = 9\cos \theta$.

- a. Limaçon with an inner loop
- b. Limaçon without an inner loop
- c. Cardioid
- d. **Circle through the pole ✓**

For $r = 9\cos \theta$: this is a circle through the pole.

28 Classify the polar curve $r = 5 + 2\sin \theta$.

- a. Cardioid
- b. Limaçon without an inner loop ✓**
- c. Circle through the pole
- d. Limaçon with an inner loop

For $r = 5 + 2\sin \theta$: this is a limaçon without an inner loop.

29 Classify the polar curve $r = 3\sin(3\theta)$.

- a. Rose with 8 petals
- b. Cardioid
- c. Rose with 3 petals ✓**
- d. Circle through the pole

For $r = 3\sin(3\theta)$: this is a rose with 3 petals.

Multiple choice — medium 15

30 A circular sector has radius 9 and central angle $\frac{\pi}{3}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

- a. 14π
- b. $\frac{27\pi}{2}$ ✓**
- c. $\frac{3\pi}{2}$
- d. 27π

$$A = \frac{1}{2}(9)^2\left(\frac{\pi}{3}\right) = \frac{1}{2}(81)\left(\frac{\pi}{3}\right) = \frac{27\pi}{2}.$$

31 A circular sector has radius 2 and central angle $\frac{2\pi}{3}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

a. $\frac{8\pi}{3}$

b. $\frac{2\pi}{3}$

c. $\frac{4\pi}{3}$ ✓

d. $\frac{11\pi}{6}$

$$A = \frac{1}{2}(2)^2\left(\frac{2\pi}{3}\right) = \frac{1}{2}(4)\left(\frac{2\pi}{3}\right) = \frac{4\pi}{3}.$$

32 A circular sector has radius 6 and central angle $\frac{\pi}{6}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

a. $\frac{7\pi}{2}$

b. $\frac{\pi}{2}$

c. 3π ✓

d. 6π

$$A = \frac{1}{2}(6)^2\left(\frac{\pi}{6}\right) = \frac{1}{2}(36)\left(\frac{\pi}{6}\right) = 3\pi.$$

33 A circular sector has radius 4 and central angle $\frac{2\pi}{3}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

a. $\frac{4\pi}{3}$

b. $\frac{16\pi}{3}$ ✓

c. $\frac{35\pi}{6}$

d. $\frac{32\pi}{3}$

$$A = \frac{1}{2}(4)^2\left(\frac{2\pi}{3}\right) = \frac{1}{2}(16)\left(\frac{2\pi}{3}\right) = \frac{16\pi}{3}.$$

34 A circular sector has radius 3 and central angle $\frac{3\pi}{2}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

a. $\frac{27\pi}{2}$

b. $\frac{9\pi}{4}$

c. $\frac{27\pi}{4}$ ✓

d. $\frac{29\pi}{4}$

$$A = \frac{1}{2}(3)^2\left(\frac{3\pi}{2}\right) = \frac{1}{2}(9)\left(\frac{3\pi}{2}\right) = \frac{27\pi}{4}.$$

35 A circular sector has radius 3 and central angle $\frac{3\pi}{4}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

a. $\frac{27\pi}{8}$ ✓

b. $\frac{27\pi}{4}$

c. $\frac{31\pi}{8}$

d. $\frac{9\pi}{8}$

$$A = \frac{1}{2}(3)^2\left(\frac{3\pi}{4}\right) = \frac{1}{2}(9)\left(\frac{3\pi}{4}\right) = \frac{27\pi}{8}.$$

36 A circular sector has radius 7 and central angle $\frac{3\pi}{4}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

a. $\frac{151\pi}{8}$

b. $\frac{147\pi}{8}$ ✓

c. $\frac{147\pi}{4}$

d. $\frac{21\pi}{8}$

$$A = \frac{1}{2}(7)^2\left(\frac{3\pi}{4}\right) = \frac{1}{2}(49)\left(\frac{3\pi}{4}\right) = \frac{147\pi}{8}.$$

37 A circular sector has radius 4 and central angle $\frac{5\pi}{6}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

a. $\frac{5\pi}{3}$

b. $\frac{20\pi}{3}$ ✓

c. $\frac{40\pi}{3}$

d. $\frac{43\pi}{6}$

$$A = \frac{1}{2}(4)^2\left(\frac{5\pi}{6}\right) = \frac{1}{2}(16)\left(\frac{5\pi}{6}\right) = \frac{20\pi}{3}.$$

38 A circular sector has radius 5 and central angle $\frac{\pi}{3}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

- a. $\frac{14\pi}{3}$
- b. $\frac{25\pi}{6}$ ✓
- c. $\frac{25\pi}{3}$
- d. $\frac{5\pi}{6}$

$$A = \frac{1}{2}(5)^2\left(\frac{\pi}{3}\right) = \frac{1}{2}(25)\left(\frac{\pi}{3}\right) = \frac{25\pi}{6}.$$

39 A circular sector has radius 9 and central angle $\frac{5\pi}{6}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

- a. $\frac{137\pi}{4}$
- b. $\frac{135\pi}{4}$ ✓
- c. $\frac{135\pi}{2}$
- d. $\frac{15\pi}{4}$

$$A = \frac{1}{2}(9)^2\left(\frac{5\pi}{6}\right) = \frac{1}{2}(81)\left(\frac{5\pi}{6}\right) = \frac{135\pi}{4}.$$

40 A circular sector has radius 5 and central angle $\frac{\pi}{2}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

- a. $\frac{25\pi}{4}$ ✓
- b. $\frac{25\pi}{2}$
- c. $\frac{27\pi}{4}$
- d. $\frac{5\pi}{4}$

$$A = \frac{1}{2}(5)^2\left(\frac{\pi}{2}\right) = \frac{1}{2}(25)\left(\frac{\pi}{2}\right) = \frac{25\pi}{4}.$$

41 A circular sector has radius 5 and central angle $\frac{\pi}{4}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

- a. $\frac{5\pi}{8}$
- b. $\frac{25\pi}{4}$
- c. $\frac{29\pi}{8}$
- d. $\frac{25\pi}{8}$ ✓

$$A = \frac{1}{2}(5)^2\left(\frac{\pi}{4}\right) = \frac{1}{2}(25)\left(\frac{\pi}{4}\right) = \frac{25\pi}{8}.$$

42 A circular sector has radius 3 and central angle π . Use $A = \frac{1}{2}r^2\theta$ to find its area.

a. 5π

b. $\frac{9\pi}{2}$ ✓

c. $\frac{3\pi}{2}$

d. 9π

$$A = \frac{1}{2}(3)^2(\pi) = \frac{1}{2}(9)(\pi) = \frac{9\pi}{2}.$$

43 A circular sector has radius 5 and central angle $\frac{5\pi}{6}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

a. $\frac{25\pi}{12}$

b. $\frac{125\pi}{6}$

c. $\frac{131\pi}{12}$

d. $\frac{125\pi}{12}$ ✓

$$A = \frac{1}{2}(5)^2\left(\frac{5\pi}{6}\right) = \frac{1}{2}(25)\left(\frac{5\pi}{6}\right) = \frac{125\pi}{12}.$$

44 A circular sector has radius 8 and central angle $\frac{\pi}{2}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

a. 16π ✓

b. 32π

c. 2π

d. $\frac{33\pi}{2}$

$$A = \frac{1}{2}(8)^2\left(\frac{\pi}{2}\right) = \frac{1}{2}(64)\left(\frac{\pi}{2}\right) = 16\pi.$$

Multiple choice — hard 15

45 Convert the polar point $(-6, \pi)$ to rectangular coordinates.

- a. $(0, 6)$
- b. $(7, 0)$
- c. $(6, 0)$ ✓
- d. $(-6, 0)$

$$x = r \cos \theta = -6 \cos(\pi) = 6. \quad y = r \sin \theta = -6 \sin(\pi) = 0. \quad \text{Rectangular point: } (6, 0).$$

46 Convert the polar point $(8, \frac{3\pi}{2})$ to rectangular coordinates.

- a. $(-8, 0)$
- b. $(0, -8)$ ✓
- c. $(0, 8)$
- d. $(0, 9)$

$$x = r \cos \theta = 8 \cos\left(\frac{3\pi}{2}\right) = 0. \quad y = r \sin \theta = 8 \sin\left(\frac{3\pi}{2}\right) = -8. \quad \text{Rectangular point: } (0, -8).$$

47 Convert the polar point $(2, 0)$ to rectangular coordinates.

- a. $(-2, 0)$
- b. $(2, 0)$ ✓
- c. $(0, 2)$
- d. $(3, 0)$

$$x = r \cos \theta = 2 \cos(0) = 2. \quad y = r \sin \theta = 2 \sin(0) = 0. \quad \text{Rectangular point: } (2, 0).$$

48 Convert the polar point $(6, \frac{3\pi}{2})$ to rectangular coordinates.

- a. $(0, -6)$ ✓
- b. $(-6, 0)$
- c. $(0, 7)$
- d. $(0, 6)$

$$x = r \cos \theta = 6 \cos\left(\frac{3\pi}{2}\right) = 0. \quad y = r \sin \theta = 6 \sin\left(\frac{3\pi}{2}\right) = -6. \quad \text{Rectangular point: } (0, -6).$$

49 Convert the polar point $(-5, \pi)$ to rectangular coordinates.

a. $(-5, 0)$

b. $(0, 5)$

c. $(6, 0)$

d. $(5, 0)$ ✓

$x = r \cos \theta = -5 \cos(\pi) = 5$. $y = r \sin \theta = -5 \sin(\pi) = 0$. Rectangular point: $(5, 0)$.

50 Find the values of θ in $[0, 2\pi)$ where the polar curves $r = 3$ and $r = 6 \cos \theta$ intersect.

a. $\theta = \frac{2\pi}{3}$ or $\theta = \frac{4\pi}{3}$

b. $\theta = \frac{\pi}{4}$ or $\theta = \frac{7\pi}{4}$

c. $\theta = \frac{\pi}{6}$ or $\theta = \frac{11\pi}{6}$

d. $\theta = \frac{\pi}{3}$ or $\theta = \frac{5\pi}{3}$ ✓

Set $k = 3 = 6 \cos \theta$: $\cos \theta = \frac{1}{2}$. This gives $\theta = \frac{\pi}{3}$ or $\theta = \frac{5\pi}{3}$.

51 Find the values of θ in $[0, 2\pi)$ where the polar curves $r = -2$ and $r = 4 \sin \theta$ intersect.

a. $\theta = \frac{\pi}{3}$ or $\theta = \frac{2\pi}{3}$

b. $\theta = \frac{7\pi}{6}$ or $\theta = \frac{11\pi}{6}$ ✓

c. $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$

d. $\theta = \frac{\pi}{4}$ or $\theta = \frac{3\pi}{4}$

Set $k = -2 = 4 \sin \theta$: $\sin \theta = -\frac{1}{2}$. This gives $\theta = \frac{7\pi}{6}$ or $\theta = \frac{11\pi}{6}$.

52 Find the values of θ in $[0, 2\pi)$ where the polar curves $r = 2$ and $r = 4\sin \theta$ intersect.

a. $\theta = \frac{\pi}{4}$ or $\theta = \frac{3\pi}{4}$

b. $\theta = \frac{\pi}{3}$ or $\theta = \frac{2\pi}{3}$

c. $\theta = \frac{7\pi}{6}$ or $\theta = \frac{11\pi}{6}$

d. $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$ ✓

Set $k = 2 = 4\sin \theta$: $\sin \theta = \frac{1}{2}$. This gives $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$.

53 Find the values of θ in $[0, 2\pi)$ where the polar curves $r = 3$ and $r = 6\sin \theta$ intersect.

a. $\theta = \frac{\pi}{3}$ or $\theta = \frac{2\pi}{3}$

b. $\theta = \frac{7\pi}{6}$ or $\theta = \frac{11\pi}{6}$

c. $\theta = \frac{\pi}{4}$ or $\theta = \frac{3\pi}{4}$

d. $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$ ✓

Set $k = 3 = 6\sin \theta$: $\sin \theta = \frac{1}{2}$. This gives $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$.

54 Find the values of θ in $[0, 2\pi)$ where the polar curves $r = 1$ and $r = 2\sin \theta$ intersect.

a. $\theta = \frac{7\pi}{6}$ or $\theta = \frac{11\pi}{6}$

b. $\theta = \frac{\pi}{4}$ or $\theta = \frac{3\pi}{4}$

c. $\theta = \frac{\pi}{3}$ or $\theta = \frac{2\pi}{3}$

d. $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$ ✓

Set $k = 1 = 2\sin \theta$: $\sin \theta = \frac{1}{2}$. This gives $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$.

55 Give an equivalent polar representation of $(3, \frac{\pi}{6})$ using a negative r -value.

a. $(-3, \frac{\pi}{6})$

b. $(3, \frac{13\pi}{6})$

c. $(3, \frac{7\pi}{6})$

d. $(-3, \frac{7\pi}{6})$ ✓

With negative r : $(-3, \frac{\pi}{6} + \pi) = (-3, \frac{7\pi}{6})$.

56 Give an equivalent polar representation of $(8, \frac{\pi}{2})$ using a negative r -value.

- a. $(8, \frac{5\pi}{2})$
- b. $(-8, \frac{3\pi}{2})$ ✓
- c. $(8, \frac{3\pi}{2})$
- d. $(-8, \frac{\pi}{2})$

With negative r : $(-8, \frac{\pi}{2} + \pi) = (-8, \frac{3\pi}{2})$.

57 Give an equivalent polar representation of $(2, \frac{\pi}{4})$ using a negative r -value.

- a. $(-2, \frac{5\pi}{4})$ ✓
- b. $(2, \frac{5\pi}{4})$
- c. $(2, \frac{9\pi}{4})$
- d. $(-2, \frac{\pi}{4})$

With negative r : $(-2, \frac{\pi}{4} + \pi) = (-2, \frac{5\pi}{4})$.

58 Give an equivalent polar representation of $(9, \frac{3\pi}{4})$ using a negative r -value.

- a. $(-9, \frac{3\pi}{4})$
- b. $(9, \frac{11\pi}{4})$
- c. $(-9, \frac{7\pi}{4})$ ✓
- d. $(9, \frac{7\pi}{4})$

With negative r : $(-9, \frac{3\pi}{4} + \pi) = (-9, \frac{7\pi}{4})$.

59 Give an equivalent polar representation of $(7, \frac{5\pi}{6})$ using a negative r -value.

a. $(7, \frac{11\pi}{6})$

b. $(-7, \frac{5\pi}{6})$

c. $(7, \frac{17\pi}{6})$

d. $(-7, \frac{11\pi}{6})$ ✓

With negative r : $(-7, \frac{5\pi}{6} + \pi) = (-7, \frac{11\pi}{6})$.